Upstream and Downstream Impacts of College Merit-Based Financial Aid for Low-Income Students: Ser Pilo Paga in Colombia*
Heterogeneity Analysis

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I. Introduction

In 2010, just 37.1% of the population between 17 and 21 years old was enrolled in tertiary education in Colombia. The poor credit market development and the lack of a extensive scholarship system were the main problems that young people face when they try to continue their studies (Mogollón & Crawford (2009)).

In particular, low-income students are the ones that face the most restrictive conditions. In addition, conditional on enrolling, the majority of low-income students attend to low quality institutions.

In 2014, the Colombian government launched a large-scale financial aid program for low-income students called Ser Pilo Paga (SPP)[1].

For being eligible for SPP, students must satisfy three conditions:

i. Merit-eligibility: Score above a cutoff in the the country's high school exit exam, SABER 11. This means score among the top 9%.

ii. Need-eligibility: Score below a cutoff in the socioeconomic index SISBEN. This index is used by the government to provide other social polices, the cutoff implies that the student must belong to the poorest 50% of households.

iii. Being admitted by a Higher education institution (HEI) with High Quality Accreditation. [2]

The authors of the original paper study the impact of removing financial constraints for low-income, high-achieving students in different dimensions. First, using a regression discontinuity design they identify the causal effect of removing financial constraints (being eligible for SPP) on enrollment and college choice. They find that for merit-eligible students, having a SISBEN index just above the

* Londoño-Vélez et al. (2020)

[1] Ser pilo paga means being smart pays off in spanish

[2] HEIs must have the Ministry of Education's Qualified Registry of minimum quality standards to provide their education services. In addition, they can apply for a certificate of High Quality Accreditation,which is a proxy for the quality of the institution.

cut-off increases the immediate probability of enrollment by 56.5% and for need-eligible students, having a test score just above the cutoff increases the immediate probability of enrollment by 86.5%. Also, the program increases the probability that low-income, high-achieving students enroll in a high-quality institution.

Second, using a difference in difference approach, the authors study the broader effects on financial aid on the entire universe of students. They find that financial aid increase college enrollment for eligible and ineligible students.

Due to data availability, my analysis is focused on the first part of the original paper. I have two goals for this project, first to identify the presence of heterogeneity in the effect of removing the financial constraints on low-income, high-achieving students by gender, age, ethnicity and employment status. Second, perform a robustness check by building a control group with the cohort immediately preceding the first that received SPP.

From a policy perspective, estimating the presence of heterogeneous effects serves to improve the evaluation of the policy, by identifying if all households respond in the same way to the relaxation of financial constraints. Regarding the robustness check, I can identify the average effect of the policy on all treated students and not only on those who are close to the cut-off point for access to treatment.

I find that the heterogeneity by employment status for need-eligible students is statistically significant; the effect of removing financial constraints in working students is lower than the effect in students that do not work. The estimates in the robustness check are in line with those of the authors.

The remainder of this paper is organized as follows. Section II describes the data. Section III presents the methodology used by the authors. Section IV describes my analysis design, Section V presents my results. Finally, Section VI concludes.

II. Data

The author focus their analysis on the first cohort of SPP; that is, students who took the SABER 11 test in 2014. To run the RD design, the authors merge administrative data from different sources. From the institution in charge of standarized testing in Colombia, Instituto Colombiano para el Fomento de la Educación Superior (ICFES), they get test scores and sociodemographic information for all SABER 11 test-takers.

SISBEN scores were provided by The Department of National Planning (DNP). From the Ministry of Education's Sistema para la Prevención de la Deserción en la Educación Superior (SPADIES), they get student observable characteristics, including enrollment status, HEI. They also use data from ICETEX, the institution that manages SPP, to identify the beneficiaries of the policy.[3]

[3]The authors use other data sources for the second part of their paper, which I do not use and not all of them are available.

The final data set of 2014 SABER 11 takers contains information on 574,269 students, of which 15,423 are eligible for SPP. Regarding the cohort I use to build the control group, it is composed of 573662 students, of which 12883 are SPP eligible. Table 1 lists the variables that I use in the estimations and provides summary statistics divided cohort and SPP eligibility.

My variable of interest is Immediate enrollment-HEI, that takes value 1 if the student enrolled in any HEI immediately after finishing high school. Except for Age and Family size, all variables are dummies that take value 1 if the student hold with that characteristic. Stratum is based on housing conditions and infrastructure, with 6 being the highest on the socio-economic scale and 1 being the lowest.

Table 1—Characterization of the 2013 and 2014 cohorts

	All				Eligible students			
	2013		2014		2013		2014	
	mean	sd	mean	sd	mean	sd	mean	sd
Immediate enrollment -HEI	0.1772	0.3818	0.1932	0.3948	0.460	0.498	0.741	0.438
Female	0.544	0.498	0.547	0.498	0.417	0.493	0.429	0.495
Age	17.790	4.291	17.910	4.488	16.300	1.298	16.610	2.313
Age-Over average	0.306	0.461	0.315	0.464	0.078	0.268	0.076	0.265
Ethnic minority	0.133	0.340	0.132	0.339	0.317	0.465	0.299	0.458
Household SES: stratum 1	0.417	0.493	0.424	0.494	0.447	0.497	0.451	0.498
Household SES: stratum 2	0.354	0.478	0.349	0.477	0.216	0.411	0.221	0.415
Household SES: stratum 3	0.180	0.384	0.177	0.382	0.017	0.129	0.023	0.151
Household SES: stratum 4	0.035	0.184	0.036	0.185	0.003	0.054	0.004	0.066
Household SES: stratum 5	0.010	0.101	0.010	0.101	0.000	0.022	0.001	0.028
Household SES: stratum 6	0.004	0.064	0.004	0.063	0.116	0.320	0.214	0.410
Mother education: primary	0.212	0.409	0.369	0.483	0.452	0.498	0.459	0.498
Mother education: secondary	0.425	0.494	0.431	0.495	0.262	0.440	0.157	0.363
Mother education: T&T	0.242	0.429	0.090	0.287	0.170	0.376	0.171	0.376
Mother education: higher	0.120	0.325	0.110	0.313	0.176	0.381	0.289	0.453
Father education: primary	0.271	0.444	0.434	0.496	0.412	0.492	0.416	0.493
Father education: secondary	0.377	0.485	0.385	0.486	0.235	0.424	0.119	0.324
Father education: T&T	0.229	0.420	0.072	0.259	0.177	0.381	0.176	0.381
Father education: higher	0.123	0.328	0.110	0.312	4.611	1.612	4.528	1.543
Family size	4.737	1.743	4.688	1.710	0.980	0.139	0.092	0.288
Employed	0.098	0.143	0.136	0.343	0.202	0.402	0.216	0.412
Private school	0.251	0.434	0.254	0.435	0.604	0.489	0.594	0.491
School hours: morning	0.516	0.500	0.513	0.500	0.006	0.076	0.008	0.088
School hours: evening	0.073	0.261	0.070	0.255	0.162	0.369	0.163	0.370
School hours: afternoon	0.159	0.366	0.156	0.363	0.004	0.064	0.007	0.081
School hours: weekends	0.058	0.234	0.065	0.246	0.061	0.238	0.102	0.303
Observations	573,662		574,269		12,883		15,423	

Source: Own calculations based on Londoño-Vélez et al. (2020)

III. Model Specification

The authors exploit SABER 11 and SISBEN cutoffs using a sharp RD design. They collapse the multidimensional discontinuity into a single dimension defining two running variables: the distance of SABER 11 score from the eligibility cutoff for those students that are need-eligible and the distance of SISBEN score from the eligibility cutoff for those students that are merit-eligible.

The causal effect identified is the impact on post-secondary enrollment of being eligible for financial aid given each of the criteria.

Let $T_i = \mathbb{1}(X_i \geq c)$ be an indicator for SPP eligibility following the criteria SABER 11 or SISBEN, where c is the point of the corresponding discontinuous assignment rule. Let $Y_i(1)$ and $Y_i(0)$, the outcomes that would be observed under the treatment (being eligible for financial aid) or control conditions respectively. In this case, Y_i is immediate enrollment in any HEI.

The observed average outcome given the score is:

$$E[Y_i \mid X_i] = \begin{cases} E[Y_i(0) \mid X_i] & \text{if } X_i < c \\ E[Y_i(1) \mid X_i] & \text{if } X_i \geq c \end{cases}$$

In the RD design, the treatment effect is defined as:

$$\tau_{SRD} \equiv E[Y_i(1) - Y_i(0) \mid X_i = c] = \mu_+ - \mu_-$$

If the functions $E[Y_i(1) \mid X_i = c]$ and $E[Y_i(0) \mid X_i = c]$ are continuos at $x = c$, the treatment effect is identified:

$$\tau_{\text{SRD}} = \lim_{x \downarrow c} E[Y_i \mid X_i = x] - \lim_{x \uparrow c} E[Y_i \mid X_i = x]$$

The authors estimate μ_+ and μ_- through a local polynomial method, that is they approximate the regression functions only near the cutoff. In particular, they use a data-driven process to find the optimal bandwidth for a polynomial of degree 1 and fit a weighted least square regression.[4]

So, $\hat{\mu}_+$ and $\hat{\mu}_-$ are defined as follow:

$$\hat{\mu}_+ : \quad \hat{Y}_i = \hat{\mu}_+ + \hat{\mu}_{+,1}(X_i - c)$$
$$\hat{\mu}_- : \quad \hat{Y}_i = \hat{\mu}_- + \hat{\mu}_{-,1}(X_i - c)$$

Valid inference should take into account the effect of misspecification. Following Calonico et al. (2014) the authors implement a bias-corrected inference procedure. The confidence intervals are defined as:

[4]The weights are determined by a triangular kernel function: $K(u) = (1 - |u|)\mathbb{1}(|u| \leq 1)$. The optimal bandwidth is determined minimizing the MSE of the local polynomial RD point estimator.

$$\mathrm{CI}_{\mathrm{rbc}} = \left[\left(\hat{\tau}_{\mathrm{SRD}} - \hat{\mathcal{B}} \right) \pm 1.96 \cdot \sqrt{\hat{\mathcal{V}}_{\mathrm{bc}}} \right]$$

Where $\hat{\mathcal{B}}$ is the estimated bias and $\hat{\mathcal{V}}_{\mathrm{bc}}$ is the estimated variance that incorporates the extra variability introduced in the bias estimation step.[5]

The authors argue that the tree key assumptions for the validity of the RD design are satisfied in their analysis. First, running variables are quasi-randomly assigned close to the SPP eligibility cutoffs. Second, any observed differences around the discontinuity is due differences in the running variables. Finally, there is a large change in assignment to treatment as a function of the running variable generated by the predicted discontinuity. They perform several tests and robustness check to validate this assumptions, the results are available mainly in the online appendix.

IV. Analysis Design

A. *Heterogeneity*

To study the presence of heterogeneity in the treatment effect, I use the same design as the authors in eight sub-samples, according four dummies variables: Female, Age-over average, Employed and Ethnic minority.

To test the presence of heterogeneity, I define the following statistic for each $j \in J$, where $J = \{Female, Age_over_average, Employed, Ethnic_minority\}$:

$$\tau_j^{DIFF} = \tau_{j=1} - \tau_{j=0} \tag{1}$$

Following the author's methodology, I report estimates separately, collapsing the discontinuity into a single dimension for each student.

I calculate the standard errors of the statistic using non-parametric bootstrap.[6]

B. *Robustness check*

Trough a RD design, we estimate the average effect around the cut-off point. In order to identify the average treatment effect on all treated students, I build a control group with the students who take the SABER-11 exam in 2013 and based on their test score and SISBEN index they would be eligible to receive treatment.

I compare the treatment group with those who would have received the treatment if born a year after. It is implicit that the only difference between both

[5] See Calonico et al. (2014) pg. 2302. for more details.

[6] From 1,000 replicates, I estimate each of the differences of interest and then compute the standard error of the 1,000 estimates.

cohorts is the students' year of birth. Table 1 (Eligible students columns) characterizes the treatment and control group through observable characteristics. The two groups are very similar in observable characteristics.

I use the following specification:

$$Y_i = \beta_0 + \beta_1 D_i + X_i'\delta + \varepsilon_i \tag{2}$$

Where Y_i is immediate enrollment in any HEI., β_1 is my coefficient of interest, represents the average difference in the immediate post-secondary enrollment there is between the two groups. X_i is a set of control variables (all those presented in Table 1) and ε_i is the error term.

V. Analysis Results

A. *Heterogeneity*

Figure 1 and 2 plots the probability of immediate enrollment in any HEI using SABER 11 and SISBEN as the running variables for each subsample of interest. The lines are fitted values from a regression of immediate post-secondary enrollment on the corresponding running variable for eligible (right of the cutoff point) and no eligible students (left of the cutoff point). The dots are cell means.

Using SABER 11 as the running variables, in all the sub-samples the figures suggest there is a discontinuous jump in the probability of immediate post-secondary enrollment at the eligibility cutoffs (Figure 1). When I separate the samples by age, employment, and minority status, I find that the discontinuity in the probability of immediate enrollment is greater when these variables take the value 0.

Using SISBEN as the running variables, in the sub-samples of females and males, there is also a discontinuous jump in the likelihood of immediate post-secondary enrollment at the eligibility cutoffs (Figure 2). In the sub-samples corresponding to the variables age, employment, and minority status taking value 0 there is a large spread between the cell means around the fitted regression, but still around the cut-off point there is a discontinuous jump in the probability of immediate post-secondary enrollment. This suggests that the effect on the total sample is due to the sub-sample with value 0 (which is also where the largest number of observations are found).

Table 2 presents the reduced-form estimates using the methodology detailed in Section III. Model Specification in each sub-sample. The first column replicate the results for all the sample presented in the original paper (Column 1 of their Table 2). In Panel A, SABER 11 is the running variables, while in Panel B, the running variable is SISBEN.

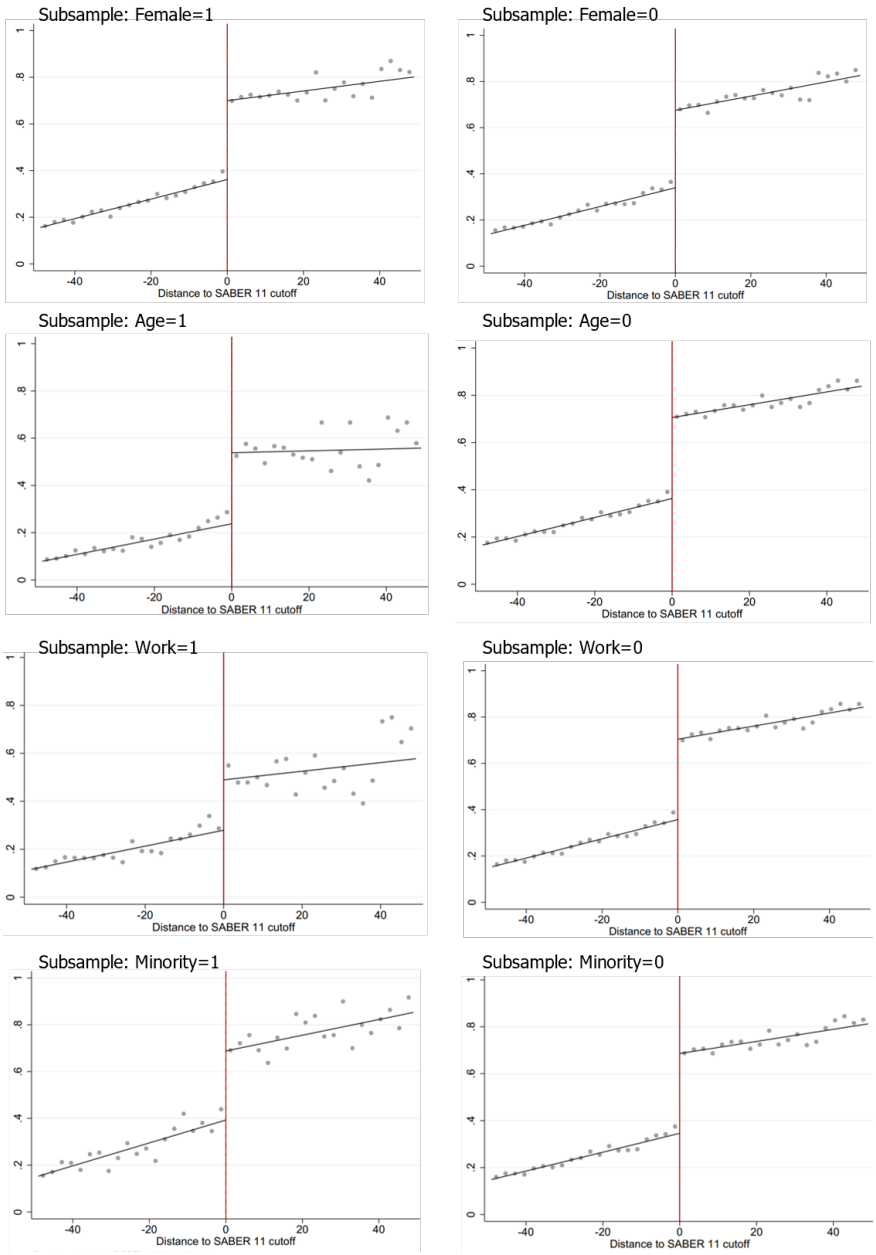

Figure 1. Immediate Postsecondary Enrollment - SISBEN eligible students

Note: The figures plot the probability of immediate enrollment in any postsecondary institution as a function of the distance to SABER 11 SPP eligibility cutoffs. Sub-samples restricted to SISBEN-eligible individuals. Sample average within bin.
Source: Own calculations based on Londoño-Vélez et al. (2020)

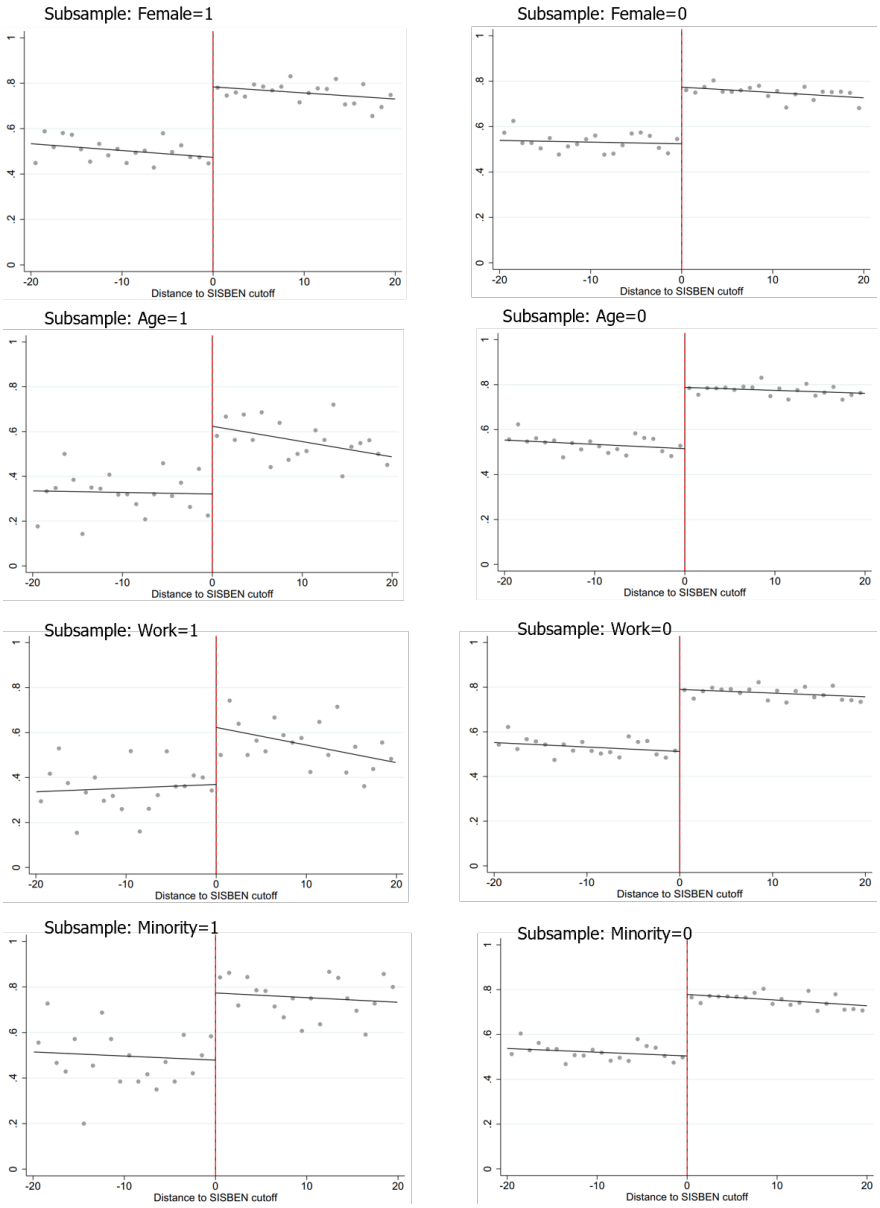

Figure 2. Immediate Postsecondary Enrollment - SABER-11 eligible students

Note: The figures plot the probability of immediate enrollment in any postsecondary institution as a function of the distance to SISBEN SPP eligibility cutoffs. Sub-samples restricted to SABER-11-eligible individuals. Sample average within bin.
Source: Own calculations based on Londoño-Vélez et al. (2020)

To interpret the magnitude of the coefficients, it is convenient to express them as a percentage with respect to the base (Mean control row in the table). For instance, using SABER 11 as running variable, relaxing the financial constraint increase the probability of immediate enrollment by 84,2% in females and 88,0% in males. When the running variable is SISBEN those effects are 73% and 45,1% respectively for females and males. As the authors express, we expect to have a larger effect using SABER 11 as a running variable, because we are estimating the effect in the sub-sample of students who are need-eligible (belong to the poorest households), so it is more likely that in the absence of the policy they would be financially constrained.

Table 2—Immediate Enrollment in Postsecondary Education - sub-samples.

Sample	All	Female	Male	Old	Young	Work	No Work	Minor.	No Minor.
Panel A. SABER 11 as the running variable									
RF	0.32	0.321	0.318	0.244	0.330	0.175	0.328	0.252	0.325
	(0.012)	(0.018)	(0.015)	(0.042)	(0.012)	(0.044)	(0.013)	(0.045)	(0.013)
Mean control	0.37	0.381	0.361	0.293	0.380	0.337	0.375	0.435	0.364
Observations	299475	172182	127293	85420	214055	40830	258645	41525	257950
BW loc. Poly.	29.679	27.434	33.501	22.072	34.205	26.827	27.422	28.243	29.523
Effect obs. Control	31170	14076	18874	2572	34537	2520	25494	2812	28244
Effect obs. Treat	11711	5098	6905	1105	11225	1010	10330	898	10803
Panel B. SISBEN as the running variable									
RF	0.274	0.323	0.238	0.337	0.269	0.232	0.276	0.274	0.277
	(0.027)	(0.042)	(0.032)	(0.082)	(0.029)	(0.099)	(0.029)	(0.117)	(0.029)
Mean control	0.485	0.438	0.528	0.308	0.500	0.368	0.494	0.592	0.476
Observations	23132	9980	13152	2129	21003	1961	21171	1557	21575
BW loc. Poly.	9.028	8.502	11.996	12.660	8.756	9.370	8.871	8.650	8.487
Effect obs. Control	3868	1577	2844	352	3525	273	3562	200	3476
Effect obs. Treat	3902	1582	2914	446	3476	323	3521	249	3391

Note: This table presents the effect of financial aid eligibility on immediate postsecondary enrollment using a regression discontinuity design in different sub-samples. Panel A uses SABER 11 test score as the running variable and restrict the sample to SISBEN-eligible students. Panel B uses SISBEN wealth index as the running variable, and restrict the sample to SABER 11-eligible students. Bias-corrected RD results estimated. Robust standard errors in parentheses.
Source: Own calculations based on Londoño-Vélez et al. (2020)

Given the small number of observations, estimates for students who are above average age, who work, and who self-identify as part of an ethnic minority have large standard errors. In any case, individually we cannot reject that these coefficients are equal to zero. Note that for the case of students below the average age, who do not work and who do not identify themselves in an ethnic minority, the treatment effect is almost equal to that of the entire population.

Table 3 presents the estimation of the difference in the treatment effect for each J (Equation 1). I calculate the standard errors of the difference in the treatment effect for each J using non-parametric boostrap. We cannot reject that the difference in the statistic by gender, age or minority status is equal to zero using SABER 11 or SISBEN as the running variable.

The difference in the effect of relaxing the financial constraint is statistically

significant by employment status when SABER 11 is the running variable. That is, the effect of having a SABER 11 score above the cut-off on students who work and are need-eligible is on average 15 percentage points lower than those who do not work and are need-eligible. One possible explanation is that for these students who are already immersed in the labor market, the opportunity cost of studying is much higher and therefore having financing to continue studying does not have as much effect.

Table 3—Heterogeneity effects on Immediate Enrollment in Postsecondary Education

	Gender	Age	Employment	Minority
Panel A. SABER 11 as the running variable				
τ^{DIFF}	0.003	-0.086	-0.153	-0.073
	(0.031)	(0.049)	(0.051)	(0.058)
Panel B. SISBEN as the running variable				
τ^{DIFF}	0.085	0.068	-0.044	-0.003
	(0.071)	(0.115)	(0.133)	(0.129)

Note: This table presents the difference on the the effect of financial aid eligibility by Gender, Age, Employment and Minority status estimated using a RD design. Non-Parametric Standard errors in parentheses, calculated after 1,000 replications.

B. Robustness check

Table 4 presents the results of model 2. The effect of financial aid eligibility estimated using as control group eligible students who took the SABER-11 in 2013 is of 30 percentage points. In absolute terms is very similar to the one estimated by the authors trough the RD design.

Table 4——Immediate Enrollment in Postsecondary Education - Robustness Check

	Model (1)	Model (2)
Effect of being eligible for financial aid	0.281	0.300
	(0.004)	(0.022)
Control variables		X
Observations	28,259	25,778

Note: This table presents the effect of financial aid eligibility on immediate postsecondary enrollment using as a control group the eligible students that took the SABER-11 one year before the implementation of the policy. The second model include the variables summarized in Table 1 as controls. Cluster standard errors by socioeconomic stratum in parentheses.

In relative terms, it represents an increase in the probability of immediate enrollment of 0.65%. As expected, the estimated effect is between the ones identified in the original paper. This is because SISBEN eligible and SABER-11 eligible

students are different: as I mentioned, students in the first group are poorer and therefore less likely to attend HEI.

VI. Conclusions

One of the main problems in the Colombian educational system was the lack of access to financing for post-secondary studies. In Londoño-Vélez et al. (2020) the authors study the effect of removing financial constraints for low-income, high-achieving students in the immediate enrollment in HEI using a RD design.

I consider the possible presence of heterogeneity effects in the treatment by gender, age, employment and ethnic minority status. I find that only the heterogeneity by employment status for need-eligible students is statistically significant. One possible interpretation is that for need-eligible students that are working while attending high-school is extremely costly study instead of working full time. This suggests that SPP was not as effective in encouraging students who were already working and who had a performance in SABER 11 in the top 9% to continue studying. Naturally, further research is necessary to understand the drivers of these students' decisions.

Finally, the robustness check shows that RD estimations are similar to the ones obtained using as control group eligible students who took the SABER-11 in 2013.

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